STUDENT'S ASSESSMENT NUMBER……

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE - REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LUDEWA DISTRICT COUNCIL
FORM TWO MOCK EXAMINATION 2022

032 **CHEMISTRY**

TIME: 2:30 Hours **8th JUNE 2022**

INSTRUCTIONS

i. This paper consists of section A, B and C
ii. Answer All questions in all sections
iii. Calculators and cellular phone are Not allowed in examination room
iv. Write your number on every page on every page on space provided
v. Constants used;-

H=1, C=12, O=16, Ca=40, Cl=35.5, K=39 ,Na=23 , N=14

FOR EXAMINERS' USE ONLY		
QUESTION NUMBER	**SCORE**	**EXAMINERS' INITIAL**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

SECTION A (15 Marks)
Answer all questions from this section

1. From item (i-x), choose the correct item from the given alternatives and write its answer beside the item number provided.

   i. A form two student was preparing different types of mixtures in the chemistry laboratory. The first mixture, he dissolved 30grams of sodium chloride crystals (NaCl) in 500cm^3 of water then he stirred the mixture for about five minutes. What is the name of this mixture prepared by a student?

   A. Heterogeneous mixture
   B. Suspension mixture
   C. Colloidal mixture
   D. Homogenous mixture

   ii. One student from form one observed that most of the kitchen tools such as plates, cups, bowl and jugs are made of steel which is smooth and shiny. As a chemistry student why these kitchen tools are made up by such steel?

   A. To make them look expensive
   B. To make them shiny
   C. To prevent them from rusting
   D. To make easy to see

   iii. Kimbe was doing experiment concerning flame test of different chemical substances in the chemistry laboratory, unfortunately his laboratory coat caught fire .Which of the following fire extinguishing items will you use to help him to extinguish fire from his laboratory coat?

   A. Fire blanket
   B. Carbon dioxide
   C. Water
   D. Form

   iv. An element H with valency of 2 react with another element L with valency of 3. What is the probable chemical formula between atoms of element H and element L?

   A. H_2L_3
   B. L_2H_3
   C. H_3L_2
   D. HL_2

   v. Jackline when was mopping chemistry laboratory she saw a sign of cross on a plastic bottle containing a certain chemical. Which of the following is the correct answer about the interpretation made by Jackline concerning the property of chemical contained in the bottle?

   A. Radioactive
   B. Harmful
   C. Toxic
   D. Oxidant

   vi. Form four students in a certain school were conducting experiment about volumetric analysis. In their experiment they were required to transfer three drops of methyl orange from the reagent bottles into the reacting vessels. Do you think which of the following apparatus they were using to transfer three drops of methyl orange from reagent bottle to reacting vessels?

   A. Beaker
   B. Burette
   C. Spatula
   D. Dropper

   vii. Students from Lugarawa health training center were doing a scientific research about malaria in Njombe region. Which among the following scientific procedures they were using to accept or to reject their hypothesis made?

   A. Conclusion
   B. Data interpretation
   C. Data analysis
   D. Experimentation

viii. During the practical session one student mixed accidentally the iron fillings with sulphur powder. Assume you are the expert of chemistry which method will you use to obtain these two components of the mixture separately?

A. Piking
B. Filtration
C. Magnetization
D. Evaporation

ix. Boron is known to have two stable isotopes of atomic masses 10.016g and 11.013g. The isotopes normally exists in nature in the proportions of 20% and 80%respectively, from the data what is the average mass of Boron?

A. 10.81g
B. 10.62g
C. 10.51g
D. 10.58g

x. Laboratory technician wanted to know the liquid present in beaker labeled L left after experiment. He took two drops of liquid L and he decided to add it into petri dish containing white anhydrous copper (II) Sulphate crystals. Immediately after the addition of three drops of liquid L into petri dish containing the anhydrous copper (II) Sulphate crystals the mixture turned to blue colour. What is the name of liquid present in beaker labeled L?

A. Water
B. Acid
C. Salts
D. Base

i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.

2. Match the items in list A with the responses in list B by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i. Means that a substance is dangerous and can cause death within a short time ii. Means that a substance is dangerous and can affect our health for long time iii. Means that the substance emits harmful radiations that penetrate human body and cause damage. iv. Means that the substance can speed up the rate of burning. v. Means that the substance can catch fire easily	A. Flammable B. Corrosive C. Explosive D. Irritant E. Toxic F. Keep away from water G. Fragile H. Oxidant I. Radio active J. Danger of electric shock K. Harmful

i.	ii.	iii.	iv.	v.

3. a) Most of farmers in the rural areas were interested to know how the knowledge of chemistry is necessary in agricultural activities. Assume you are a chemist; educate these farmers on the application of chemistry in agriculture by giving five points.

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ii. ……
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iii. ……
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iv. ……
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b) The headmaster of a certain school was bought a small box which had a sign of Red Cross. Most of form one students they did not have a knowledge about that box.by using knowledge that you have learnt from chemistry answer the following questions.

i. What is the name given to that box

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ii. Briefly explain the functions of four possible items that can be found in a box

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4. a) A laboratory technician advised students to use non-luminous flame when they are heating chemicals in the laboratory. As a form two student explain briefly four reasons why he encouraged students to use this kind flame?

i. ……
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ii. ……
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iii. …….....
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b) Some people in the society believe that solving problems by using scientific procedure is the wastage of time and money. By using knowledge of chemistry that you have refute the statement by giving four points.

i. ……
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ii. ……
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iii. ……
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iv. ……
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5. a) Assume you are given two gas jars, one gas jar containing gas A and another gas jar containing gas B. Gas A is used to fill balloons while gas be is used to by ocean deep divers.

i. What are the chemical test of gas A and gas B?

Gas A……..

Gas B……..

ii. Explain five uses of gas A in your daily life.

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b) Mwahisa placed a yellow solid on a deflagrating spoon, ignited and then lowed the spoon into a gas jar of oxygen .The solid burnt vigorously with a yellow flame and finally the product was a pale yellow solid.

i. What is the name of the yellow solid that is burnt in oxygen

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ii. Write the chemical formula of the pale yellow product formed

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iii. Write the word equation for the reaction

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6. a) During the rains many rivers fill with muddy water, which people use for drinking. This water has caused them to suffer from water borne diseases such as typhoid and diarrhea. Explain two simple methods that can be used to obtain clean water from these rivers muddy water.

i. ……

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ii. ……

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b) One form one student from a certain school was given a task by her teacher to write the chemical symbols of the following elements; sodium, iron and chlorine. But the student did as shown below in 6b (i, ii, and iii). Help him by identifying his mistakes and then write a correct symbol in each

i. Symbol of sodium is (So)

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ii. Symbol of iron is (Ir)

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iii. Symbol of chlorine is (CL)

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7. a) A form one student from a certain school given a task by her teacher to state the type of fire extinguisher(s) that she will use to extinguish fire caused by the burning materials listed below. Unfortunately a student failed to attempt this question correctly. As a form two student help her to answer this question.
   i. Burning curtain
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   ii. A fire started by a short circuit in an electric iron
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   iii. A fire started by hot oil in a frying pan
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   iv. Fire started on a bucket containing sodium metal
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   b) Cars, motorcycles and corrugated iron sheets in coastal towns like Dar es Salaam, Tanga and Zanzibar are found to rust more quickly than in other cities in Tanzania like Songea, Njombe, Mwanza and Arusha. Why is this so?
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8. a) A compound consists of 27% sodium, 16.5% nitrogen and 56.5% oxygen by mass. Calculate the simplest formula of the compound.
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b) During the practical session, Asha found some bottles in the laboratory containing different chemical compounds with different chemical formulas, unfortunately she failed to write the IUPAC names of these chemical compounds. As form two student help her by writing the correct IUPAC names of these chemical compounds found by Asha on bottles in the laboratory.

i. KCl…….
ii. Na_2SO_4 .....
iii. $CaCl_2$……..
iv. $AlCl_3$……...
v. SF_6……

9. a) Oxygen exists naturally in three different isotopes which are, ${}^{16}_{8}O$(99.76%), ${}^{17}_{8}O$(0.04%) and ${}^{18}_{8}O$(0.20%). Calculate the relative atomic mass of oxygen

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b) Use the knowledge that you have complete the following table below by indicating the number of protons, electrons and neutrons of different atoms shown in the table below. The atomic number and mass number are given

No	Atom	Atomic number	Mass number	Protons	Electrons	Neutrons
i.	Beryllium	4	9			
ii.	Fluorine	9	19			
iii.	Sodium	11	23			
iv	Oxygen	8	16			

SECTION C: 15 MARKS

10. Nowadays the world is suffering from climatic changes which is caused by over dependent on forests as a source of fuels, and over production of waste materials that pollute our environment. To overcome this problem is through the use of renewable source of energy such as biogas produced from animal dungs and sewage wastes. As the expert of chemistry prepare a report to convince the community to use this gas produced by animal dungs as the alternative source of energy while they are conserving the environment.

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